

Osmotic power: a fresh look at an old experiment

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ABSTRACT Electricity from osmotic pressure might seem a far-fetched idea but this article describes a prototype in Norway where the osmotic pressure generated between salt and fresh water drives a turbine. This idea was applied in a student investigation, where they were tasked with researching which alternative materials could be used for the semipermeable membrane and how advances in nanotechnology could provide a suitable material allowing full-scale power stations to be built. This link between nanomaterials and large-scale power production came as a surprise to the student group.

Occasionally you will find a completely new reason to perform an old experiment. My own example of rediscovery followed a trip to the visitor centre at the Statkraft hydroelectric power plant in mid-Wales, where a series of display boards detailed the concept of electrical power generation by osmosis. Subsequently, I discovered that this technology was featured in the BBC (2009) television programme *Coast*. Osmosis occurs owing to the random motion of particles wherever two solutions of different concentrations meet at a semipermeable membrane. The spontaneous passage of water from dilute to concentrated solutions through a membrane generates a pressure difference that can be harnessed to generate power. In the case of a Statkraft prototype plant sited on the banks of the Oslofjord in southern Norway, the power arises from mixing fresh and salt water. The two liquids are pumped to either side of a membrane, where osmosis creates a pressure equivalent to a column of water 120 m high. This pressure

drives a turbine to generate electricity. The technology is clearly illustrated in a short video clip (Statkraft, 2011).

The physical mechanism behind this approach can be demonstrated easily using a standard school biology experiment (Figure 1). A small bag made from Visking membrane is filled with either sugar or salt solution and attached to the bottom of a tall capillary tube. The bag is then immersed in a beaker of tap water. Full details of this experiment can be found in many school and college science books (for example, Whitten, Gailey and Davis (1988)).



Figure 1 The experimental set-up

The experiment demonstrates how the addition of either of these solutes to water forms a solution with a lower vapour pressure than that of water alone. When immersed, the solution is separated from the water by a semipermeable membrane. Water molecules will pass through the membrane into the solution in order to further dilute it and raise its vapour pressure back to its original; that is, that of water. The resulting osmotic pressure that is established by this passage of solvent molecules generates a hydrostatic head within the solution, which causes it to climb the capillary. Dyeing the sugar or salt solution with, for example, methylene blue, allows its progress up the capillary to be easily viewed.

A group of my students performed this experiment as part of a science club project on nanomaterials, since the design of a viable porous membrane is perhaps the single greatest materials challenge facing engineers and scientists hoping to industrialise this technology. I began by showing students the video clip of the technology (Statkraft, 2011), and then I asked them to discuss the challenges facing the designers of the membrane. Student suggestions included:

- knowing the precise dimensions of water and salt molecules;
- manufacturing materials with pores of appropriate dimensions;
- manufacturing materials that were thin but tough and that could also survive in a corrosive saline environment.

We then set up two experiments, one using sugar solution and the other using salt solution. Our experimental set-up is shown in Figure 1.

We found that, in both cases, the coloured solutions rose rapidly up the capillaries until they started to spill from the tops of the tubes.

The students were then tasked with researching whether Visking was suitable for use in osmotic power stations, which alternative materials could be used, and how advances in nanotechnology could lead to the development of materials such as graphene that might one day provide a suitable membrane allowing full-scale power stations to be built. This link between nanomaterials and large-scale power production came as a surprise to the student group. According to Statkraft, the company behind the prototype plant, success in this project could have important implications for the future of electrical power production. An article in the magazine *Making It: Industry for Development* (United Nations Industrial Development Organisation, 2010) said:

Many of the world's major cities are on river estuaries where sea and fresh water are easily available, and Statkraft estimates the total global potential of osmotic power to be around 1700 terawatt-hours per year – about 10% of the world's current electricity consumption ... Unlike wind and solar power, osmotic power can provide a continuous source of energy, although seasonal river-level changes do cause some fluctuations.

Whatever the future holds for osmotic power generation, and it seems that Statkraft has for the moment abandoned this line of research (Statkraft, 2013), it is certainly a technology that more science and engineering students – our future energy problem solvers – should be thinking about.

Acknowledgements

The author would like to thank the students and staff of the Mond Centre for Science and Maths at Cronton Sixth Form College; in particular, thanks go to Linda Wright, Senior Science Technician.

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